Protective effect of zingerone, a dietary compound against radiation induced genetic damage and apoptosis in human lymphocytes.

Zingerone a dietary compound was investigated for its ability to protect against radiation induced genotoxicity and apoptosis in human lymphocytes growing in vitro. The radiation antagonistic potential of zingerone was assessed by alkaline comet, cytokinesis-block micronucleus, apoptosis and reactive oxygen species inhibition assays. Treatment of lymphocytes with zingerone (10μg/ml) prior exposure to 2Gy gamma radiation resulted in a significant reduction of frequency of micronuclei as compared to the control set of cells evaluated by cytokinesis blocked micronucleus assay. Similarly, treatment of lymphocytes with zingerone prior to radiation exposure showed significant decrease in the DNA damage as assessed by comet parameters, such as percent tail DNA and Olive tail moment. Further, treatment with zingerone (10μg/ml) before irradiation significantly decreased the percentage of apoptotic cells analyzed microscopically method and by DNA ladder assay. Similarly, the radiation induced reactive oxygen species levels were significantly ($P<0.01$) inhibited by zingerone. Our study demonstrates the protective effect of zingerone against radiation induced DNA damage and antiapoptotic effect in human lymphocytes, which may be partly attributed to scavenging of radiation induced free radicals and also by the inhibition of radiation induced oxidative stress.